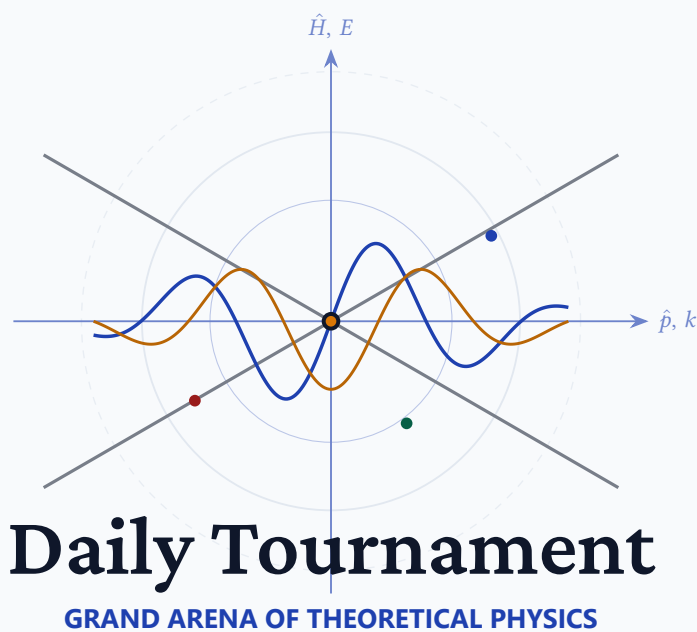




Daily Tournament

GRAND ARENA OF THEORETICAL PHYSICS

“Today’s contestants are ready: 604 Calculus $\times$ 811 Quantum Mechanics”

Closed-book · Pen-and-paper derivation · No shortcuts

Date	August 31, 2026 Week 1 (Day 1)
Modules	604: Limits, Continuity & Series Expansion 811: 1-D Potentials & Superposition
Error Protocol	5-Factor Attribution: Knowledge Gap Modelling Logic Calculation Time
Target	Standard 5.5–6.5 h Minimum Defence 3.5 h

Theoretical Physics Training · Daily Tournament

August 31, 2026 | Week 1 (Day 1) | Phase: Foundation Diagnostics & Potential Wells

§1 Tactical Briefing & Warm-Up Retrieval

- **604 Calculus** — Master the boundary conditions for equivalent infinitesimals; complete one multi-order Taylor expansion limit problem (M1).
- **811 Quantum Mechanics** — Retrieve the infinite square well boundary conditions, normalisation, and energy spectrum from memory; complete one superposition state evolution problem (Q1).
- **Supporting subjects** — English: one reading comprehension passage with sentence analysis; Politics: 30 min on Marxist philosophy fundamentals.

Momentum Defence Line: Standard push 5.5–6.5 h | Minimum defence 3.5 h (must secure: 811 retrieval + M1 + English reading).

Warm-Up Retrieval (5–10 min): Without notes, write down from memory: $\square \sin x \sim x$, $\square \ln(1+x) \sim x$,
 $\square E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$

§2 Core Problem Set

Complete independently under timed conditions

Problem M1 | 604 Calculus: Equivalent Infinitesimals & Limits

Suggested time: 25 min

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\ln(1 + \sin x) - \sin x + \frac{x^2}{2}}{x^3}.$$

Requirements: Clearly state the order of expansion retained and provide rigorous justification for each step.

Problem Q1 | 811 Quantum Mechanics: Infinite Well Superposition

Suggested time: 35 min

A particle is confined in a 1-D infinite square well on $0 < x < L$ with eigenstates $u_n(x) = \sqrt{2/L} \sin(n\pi x/L)$ and energies $E_n = n^2 \pi^2 \hbar^2 / (2mL^2)$.

The initial state is $\psi(x, 0) = \frac{1}{\sqrt{2}} [u_1(x) + u_2(x)]$.

- Find the time-dependent wave function $\psi(x, t)$.
- Compute the energy expectation value $\langle H \rangle$.
- Derive the position expectation value $\langle x \rangle_t$ and identify the beat-frequency oscillation.

Extension | Conceptual Analysis

Optional · Suggested time: 10 min

If $f(x) \sim g(x)$ as $x \rightarrow 0$, does it necessarily follow that $f(x) + x^2 \sim g(x) + x^2$? Justify your answer with proof or counterexample.

§3 Model Solutions & Commentary

Complete your independent attempt on scratch paper before consulting this page.

M1 · Model Solution & Commentary

Strategy: The denominator is x^3 , so the numerator must be expanded to $O(x^3)$; first-order terms cancel, requiring careful treatment of cross-terms in composite expansions.

Derivation: Let $y = \sin x = x - \frac{x^3}{6} + o(x^3)$. From $\ln(1+y) = y - \frac{y^2}{2} + \frac{y^3}{3} + o(y^3)$:

$$\left(\sin x - \frac{\sin^2 x}{2} + \frac{\sin^3 x}{3} + o(x^3) \right) - \sin x + \frac{x^2}{2} = -\frac{\sin^2 x}{2} + \frac{\sin^3 x}{3} + \frac{x^2}{2} + o(x^3).$$

Since $\sin^2 x = x^2 - \frac{x^4}{3} + o(x^4)$, the difference with x^2 is $O(x^4)$ (no x^3 contribution), while $\sin^3 x = x^3 + o(x^3)$. The leading third-order term is $\frac{x^3}{3} + o(x^3)$, giving:

$$\lim_{x \rightarrow 0} \frac{\ln(1 + \sin x) - \sin x + \frac{x^2}{2}}{x^3} = \frac{1}{3}.$$

[Common Error] Never use $\ln(1+\sin x) \sim \sin x$ here — the first-order terms cancel against later terms, so expansion must be carried to the highest non-cancelling order.

Q1 · Model Solution & Commentary

Physical Picture: A superposition of energy eigenstates exhibits non-trivial time evolution; the position expectation value oscillates at the Bohr frequency $\omega_{21} = (E_2 - E_1)/\hbar$ — a quantum beat.

Derivation:

1. Time-dependent wave function: $\psi(x, t) = \frac{1}{\sqrt{2}} [u_1(x) e^{-iE_1 t/\hbar} + u_2(x) e^{-iE_2 t/\hbar}]$.
2. Energy expectation value: $\langle H \rangle = \frac{1}{2}(E_1 + E_2) = \frac{5\pi^2 \hbar^2}{4mL^2}$.
3. Position expectation: $\langle x \rangle_t = \int_0^L x |\psi(x, t)|^2 dx$. Diagonal elements give $\langle 1|x|1 \rangle = \langle 2|x|2 \rangle = L/2$; the cross matrix element is:

$$\langle 1|x|2 \rangle = \frac{2}{L} \int_0^L x \sin \frac{\pi x}{L} \sin \frac{2\pi x}{L} dx = -\frac{16L}{9\pi^2}.$$

Combining: $\langle x \rangle_t = \frac{1}{2} \langle 1|x|1 \rangle + \frac{1}{2} \langle 2|x|2 \rangle + \text{Re}[\langle 1|x|2 \rangle e^{-i(E_2 - E_1)t/\hbar}]$, hence:

$$\langle x \rangle_t = \frac{L}{2} - \frac{16L}{9\pi^2} \cos \frac{(E_2 - E_1)t}{\hbar}.$$

[Checkpoint] Cross-terms produce cosine upon taking the real part of complex conjugate pairs. Verify: initial normalisation coefficient $1/\sqrt{2}$ is not dropped; dimensional check yields length scale L .

§4 Daily Debrief & Reflection

Post-Match Review (fill in before sleep, 5 min)**604 Calculus**Time spent: _____ min | M1 status: ☐ Independent ☐ Used hints ☐ IncompleteError type: ☐ Knowledge gap ☐ Modelling ☐ Logic ☐ Calculation ☐ Time**811 QM**Time spent: _____ min | Q1 status: ☐ Independent ☐ Used hints ☐ IncompleteError type: ☐ Knowledge gap ☐ Physical model ☐ Operator algebra ☐ Calculation ☐ Time**Planning****Spaced retrieval in 3 days:** _____**Today's assessment:** ☐ Peak performance ☐ Steady progress ☐ Minimum defence mode